

Section 4E. Hypothesis Testing in Logistic Regression

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Hypothesis Testing in Logistic Regression (single variable)

How do we decide if the input influences the output? We can use the hypothesis testing framework...

1. If $\beta_1 = 0$ (i.e., the input does not influence the output), then $\hat{\beta}_1 \sim \mathcal{N}(0, \text{SD}(\hat{\beta}_1)^2)$
2. Define the variable $z_1 = \hat{\beta}_1 / \text{SD}(\hat{\beta}_1)$. Hence, we have that (see Hypothesis Testing section)

$$\Pr(|z_1| < 2) = 0.95$$

3. Therefore, if $|z_1| > 2$, we have statistical evidence to reject the hypothesis that the input does not influence the output.

[Credit: James et al., ISL book]	Coefficient	Std. Error	Z-statistic
Intercept	-10.6513	0.3612	-29.5
balance	0.0055	0.0002	24.9

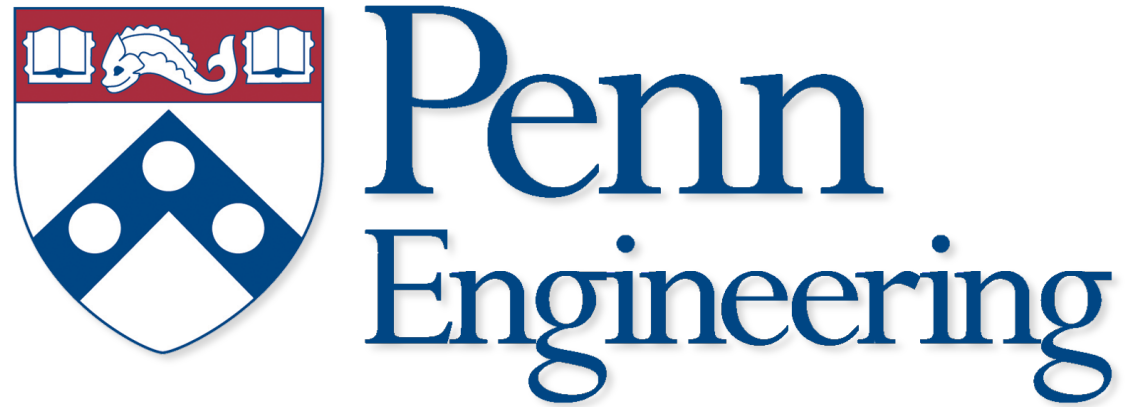
Hypothesis Testing with Multiple Variables

When $p > 1$ we can use a *multivariate logistic function*, defined below:

$$p_1(\mathbf{x}; \beta_0, \beta_1, \dots, \beta_p) = \frac{e^{\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p}}{1 + e^{\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p}}$$

- ▶ We can find estimates $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ using the Maximum Likelihood Criterion
- ▶ Predictions are made using the function $\hat{p}_1(\mathbf{x}) = p_1(\mathbf{x}; \hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$
- ▶ We can decide if a particular input is not relevant looking at the Z-values

[Credit: James et al., ISL book]	Coefficient	Std. Error	Z-statistic
Intercept	-10.8690	0.4923	-22.08
balance	0.0057	0.0002	24.74
income	0.0030	0.0082	0.37
student [Yes]	-0.6468	0.2362	-2.74



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